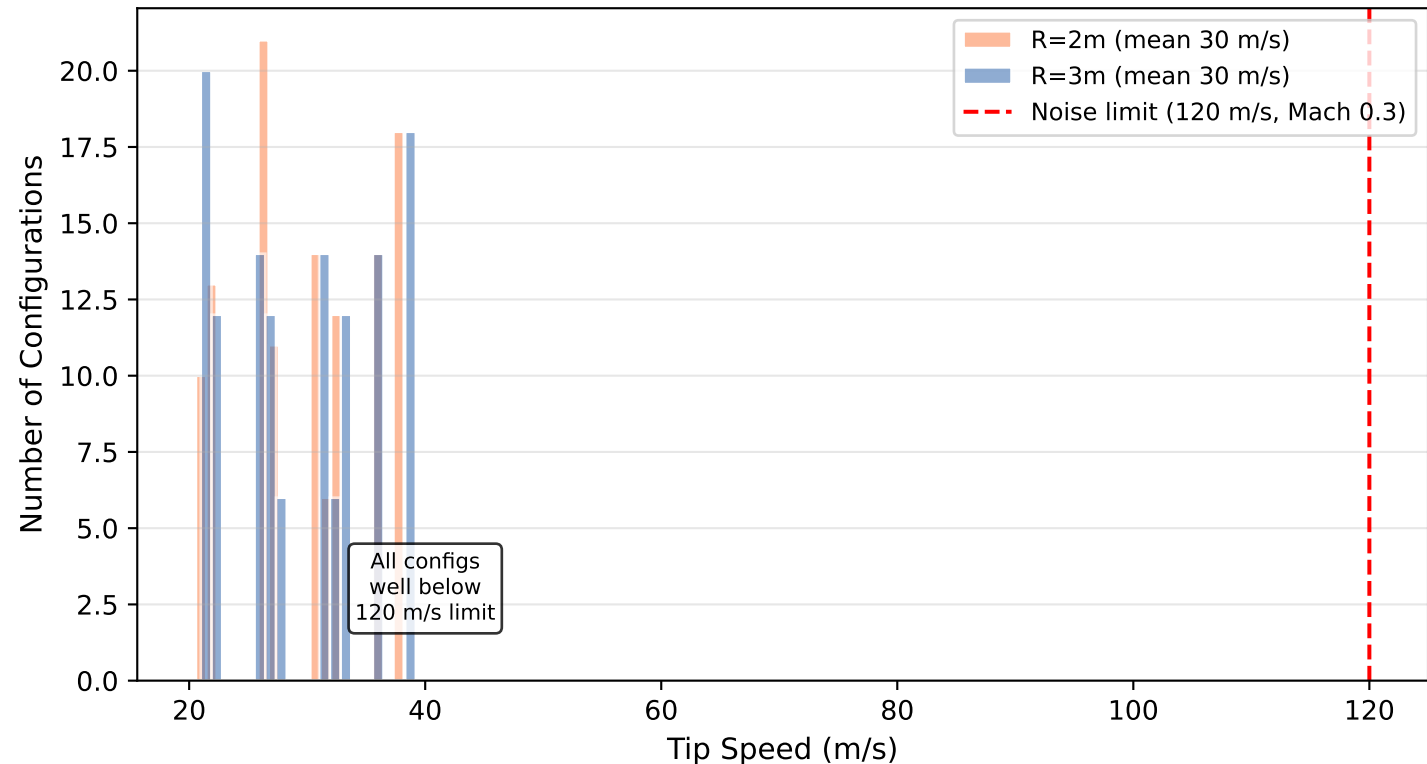


Tip Speed Distribution by Rotor Radius BEM v2.1 — Viable Configurations



IMPLICATIONS: Tip speeds range from 2–40 m/s across all viable configurations — well below the 120 m/s (Mach 0.3) noise limit. R=3.0m rotors run at 30 m/s mean tip speed. The autorotation RPM is self-limiting: higher wind increases RPM but also increases through-disk velocity, keeping tip-speed ratio roughly constant. Noise is not a constraint for the lifting autogyro configuration. Data: BEM v2.1, all wind speeds 6–12 m/s.